Adversarial Cooperation

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Chapter 1

Trust Establishment

Abstract. In cooperative systems the parties make no assumptions about channel security; this stage is therefore called *trust establishment*. Classical *unauthenticated* key-exchange schemes provide confidentiality against passive eavesdroppers, yet they fall to active man-in-the-middle attacks. Cooperation cannot rely on unverified assumptions. Our sole assumption is access to a local true-random generator; parties remain vulnerable only to signal-loss.

We propose that each participant first isolates and safeguards an experiment to measure a truly random source, then models the resulting noise according to their individual preferences. No harm arises if other parties learn one another's secret keys, secrets are tactical disadvantages.

1.1 Topic

Let Λ model n raw entropy sources Π_n, where λ^i are the *eigenvalues* of the covariance.

$$\mathcal{N} := (-1,1) \subset \mathbb{R}, \qquad \mathcal{N}_p = \mathcal{N} \mod p,$$

$$\substack{\lambda_i \\ \mathcal{N}_p} \Lambda^{\Pi_n},$$

$$k_{i,n} = \Lambda_i\big(\Pi_n\big)\big|_{\mathcal{N}_p}.$$

$$\mathcal{N} := (-1,1) \subset \mathbb{R}, \qquad \mathcal{N}_p := Q_p\big[\mathcal{N}\big] \subset \mathbb{Z}_p, \quad Q_p(x) = \big\lfloor (x+1)\tfrac{p-1}{2} \big\rfloor \pmod{p}.$$

Protocol 1: Private-key derivation from a random signal.

Algorithm 1: Entropy-Maximised Private-Key Derivation ($\mathsf{Alt}\,1$)

Input: Raw entropy batch Π_n, participant index i, target prime p
Output: Private key k_n^i

Isolate local entropy experiment `// Physical shielding`
$\pi \leftarrow \textsc{MeasureRandomSource}(\Pi_n)$
$\Lambda^i \leftarrow \textsc{SolveMaxEntropyModel}\big(\pi, \text{constraints}_i\big)$
$\mathcal{N}_p \leftarrow \textsc{OptimiseQuantiser}\big(\Lambda^i, p\big)$
$u \leftarrow \Lambda^i(\pi)$ `// continuous` $(0, 1)$ `value`
$k_n^i \leftarrow \lfloor u \cdot p \rfloor \bmod p$
$k_n^i \leftarrow \textsc{ExtractUniformBits}\big(k_n^i\big)$ `// e.g. HKDF`
return k_n^i

Step **??** of Algorithm **??** embeds user preferences into Λ^i.

Appendix Title

Details relevant to the appendix.